Design and Implementation of Three-Phase Grid-Connected Inverter for PV System

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***Abstract*— Solar energy is one of the renewable energy that has a large potential in Indonesia. Solar energy can be converted into electrical energy using PV. According to the increasing energy demand in Indonesia and the enactment of the feed-in tariff by the government, it will be easier for the sale and purchase of electricity through the on-grid system. This study discusses the design and implementation of a three-phase grid-connected inverter. This three-phase inverter circuit uses sinusoidal pulse width modulation for input signal (SPWM) is used to input the IR2113 circuit a three-phase inverter driver. SPWM used is also a synchronization method to connect the inverter to the grid. Grid synchronization in this study is equating the phase angle and inverter output frequency to the grid. In this design add DC to DC converter type push-pull converter as a three-phase inverter supply and also as a control to adjust the inverter output voltage to fit the grid. The push-pull converter uses an SG3525 PWM IC generator. The circuit can be adjusted according to the voltage you want to produce. The circuit uses a high-frequency transformer as a step up the voltage. The result of the system design is the PV output voltage is increased by the push-pull converter and then to supply three-phase inverter.**

Keywords—SPWM, three-phase inverter, push-pull converter, grid synchronization.

I. INTRODUCTION

At present, renewable energy is a very popular topic of study in the world. Solar energy is one of renewable energy which is very abundant, eco-friendly, and easy to implement. Based on data from the National Energy Council the potential of solar energy in Indonesia is an average of 4.80 kWh/m2/day [1]. Solar energy can be used directly to meet electricity needs using PV (photovoltaic) modules.

Research on the implementation of solar energy is growing very rapidly at this time. One example is the use of PV to meet electrical energy in buildings. In general, PV requires several power electronic circuits such as DC-DC converters and inverters. DC-DC boost converter equipped with PI control is used to increase the input voltage to reach DC-link voltage, then the DC link voltage is converted into AC voltage using an inverter [2]. One of the switching in the inverter techniques is SPWM which is built by comparing sine waves with triangle signals [3]. SPWM switching techniques can also be generated digitally using lookup tables on microcontroller programs [4].

In general, on-grid inverters use a DC-DC converter to increase the PV voltage to match the DC voltage of the link. Among several types of DC-DC converters that are often used are boost converters. But in fact, the input voltage required by a three-phase inverter is very high. And the boost converter output voltage is only affected by the duty cycle. So the boost converter is difficult to reach the voltage needed by the inverter if the PV voltage differs considerably from the DC link voltage. Below is a three-phase inverter scheme using a boost converter [5].

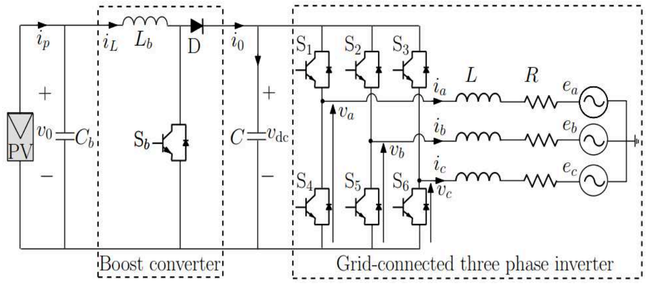

Fig. 1. Previous research three grid connected.

In this study, we designed a grid-connected three-phase inverter in a photovoltaic system. In this system, a push-pull converter is added which is used to increase PV voltage. The push-pull converter output voltage is controlled in a closed loop. The advantage of this system is that it can produce large voltages with smaller dimensions and output voltage can be controlled according to the voltage used. The grid connection system uses the SPWM method because it has a robust and easy to implement the circuit [6].

Similar research has been carried out using a cascaded H-bridge multilevel inverter for large-scale solar power plant applications [7]. While for small-scale solar power plant, 3 phase inverters are more recommended to be applied because 3 phase inverters have easier switching techniques.

Many reference methods exist regarding grid synchronization applied to an inverter. These methods include zero crossing detection (ZCD), Kalman Filters, Discrete Fourier transform (DFT), Nonlinear Least Square (NLS), Adaptive Notch Filtering (ANF), Artificial Intelligence (AI), Delayed Signal Cancellation (DSC), Phase Locked Loop (PLL), and Frequency Locked Loop (FLL), which each has a different method and analysis [8]. In general, the grid synchronization method is illustrated through the diagram below.

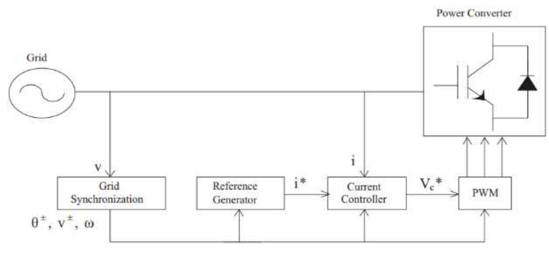

Fig. 2. General block diagram grid synchronization.

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The commonly used method is the PLL (phase locked loop) method. This method has 3 control components, namely, a phase detector (PD), a loop filter (LP), and voltage-controlled oscillator (VCO). The PLL method structure is feedback control that automatically adjusts the phase of the local signal generation to adjust the input signal phase. This method is very easy to adjust because the construction of waves takes references directly through the grid without any wave synthesis.

II. METHODOLOGY

At present, PV (photovoltaic) use with grid-connected systems is more popular than stand alone. Grid-connected systems are more cost-effective because they don't need batteries but only as an option. By using of grid-connected PV systems utilizes a tool that is capable of sending or flowing currents into the grid network, so that the power used in the grid for loading will be lighter or reduced. The tool is an on-grid inverter. Most on-grid inverters used are 1-phase on-grid inverters.

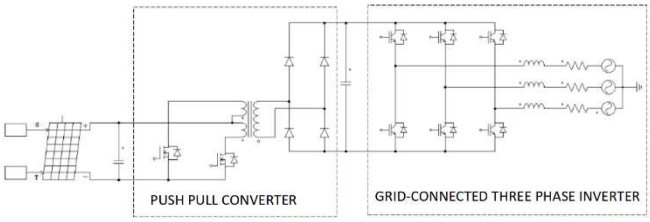


Fig. 3. Purposed configuration system.

In this study, we designed a grid-connected 3 phase inverter in a PV system to be used for buildings that have 3 phase networks. The concept of an on-grid inverter is almost the same as an off-grid inverter, but the sine wave generated at the AC voltage that is issued must be the same as the wave that is owned by the grid in order for network synchronization to occur. Grid synchronization has several variables that must be conditioned namely voltage, frequency, and phase angle at the inverter output must be the same as the grid used. Figure 1 shows a block diagram of the system that we designed.

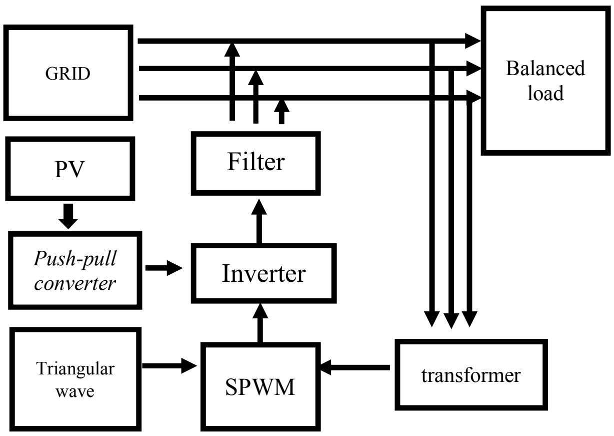


Fig. 4. Block diagram of three phase solar PV inverter on–grid.

The PV module used in this study has a maximum power of 180 Wp, a maximum voltage of 35.4 V, and a maximum current of 5.09 A. The PV specification used is obtained from testing the Standard Test Condition (STC). The design of this system uses two PVs arranged in series. Taking into account solar radiation and temperature which cannot reach ideal conditions in testing, so the PV output voltage used as a parameter is 60 V.

A. Push-Pull Converter

A DC-DC converter is used to supply the voltage needed by a 3-phase inverter. In this study, the DC-DC converter used is a push-pull converter topology, between the input side and the output isolated by the transformer. The push-pull converter output voltage can be varied by adjusting the variation of the duty cycle. The basic push-pull converter consists of two switches (transistors), transformers, rectifier diodes and filters like Figure 5.

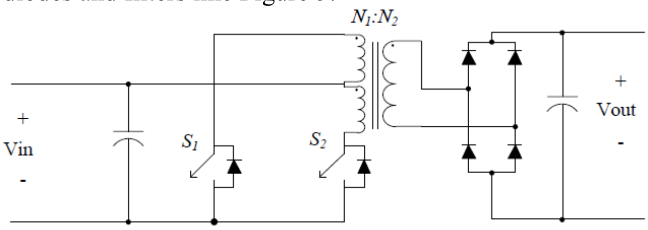


Fig. 5. Push-Pull Converter.

The primary side of the transformer is supplied alternately from S1 and S2. So as to produce alternating current towards the transformer. The output voltage on the transformer secondary is rectified using a diode. In the design, we use a full wave rectifier with four diodes, so it does not require a center tap (CT) on the transformer secondary side. The capacitor filter is used so that the push-pull converter output voltage becomes pure DC.

The expected inverter output voltage is 380 Vrms. In order to achieve this value, the inverter requires a DC voltage of $380\sqrt{2}$ which is close to 538 volts. So, the push-pull converter is designed to convert 60 V to 538 V. Equation (1) is used to determine the push-pull converter output voltage,

$$Vout = 2\frac{N2}{N1}D.Vin \tag{1}$$

To avoid voltage drop, the output voltage in the design is made greater than the expected output voltage. Based on the calculation, the coil push-pull transformer ratio is 7 + 7 winding for primer and 80 winding for secondary. The switching frequency used for push-pull converters is 40 KHz. Because both transistors must be switched alternately, the duty cycle used should not exceed 0.5. The voltage-mode PWM SG3525 module is used to adjust the duty cycle. Besides that, it can also be used to keep the output voltage constant [9]. The output voltage is controlled in a closed loop and the desired voltage is adjusted using a voltage dividing resistor. PWM control configuration on the push-pull converter equipped with feedback as shown below.

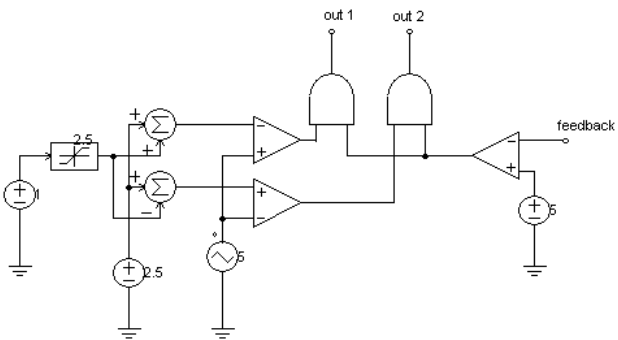


Fig. 6. PWM Push-Pull Converter Controller.

B. Three-Phase Inverter

The inverter is a circuit used to convert DC voltage to AC. One of the most common types of inverters is voltage source inverter (VSI). Three phase inverters are widely used for equipment that requires high power. The basic three-phase inverter circuit consists of 6 switches as shown in Figure 7. The characteristics of the corresponding electronic components as switches must have a high power rating and can be used for switching at high frequencies. So that the Insulated Gate Bipolar Transistor (IGBT) was chosen for the three-phase inverter design.

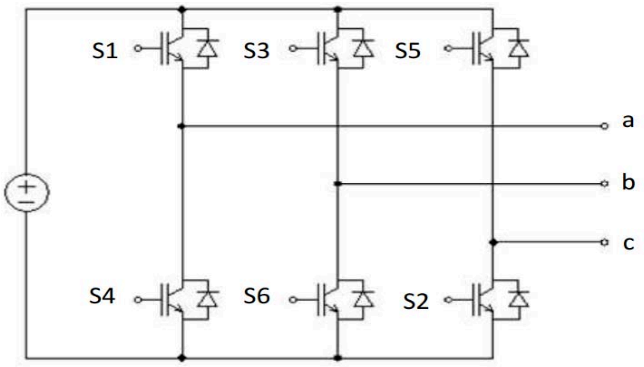
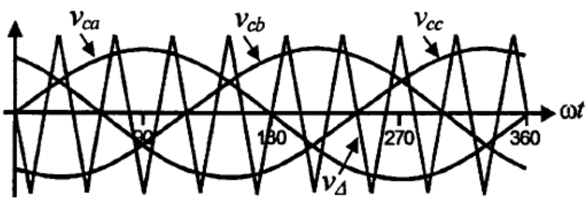


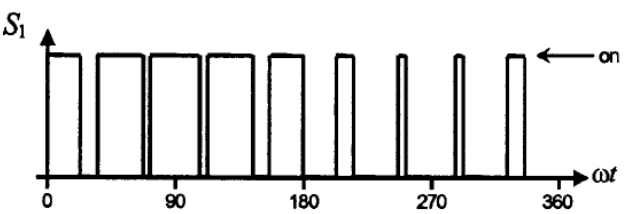
Fig. 7. Three Phase Inverter.

C. SPWM (Sinusoidal Pulse Width Modulation)

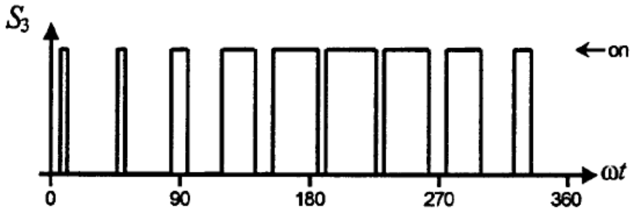
In the inverter, the process of forming a sine wave at the inverter output occurs due to the switching of the MOSFET / IGBT [10]. This switching forms a sinusoidal signal. The SPWM method is one of the switching methods that is often used to produce inverter output that forms sinusoidal signals. In this study, the SPWM method is also used as a grid synchronization control method. SPWM is built from a comparison of the sine (AC) signal with the carrier signal (triangle wave) which will produce the SPWM signal [11]. In order for the synchronization to occur, the first thing that must be considered is the phase angle and frequency that the grid has. While the voltage will be controlled by a DC-DC converter. To be able to produce a sine signal that is the same as a grid, a sine signal that is compared is a signal taken directly from the grid. In order for the sinus signal to be read by the IC TL074 op-amp component, the grid voltage is lowered to 3VAC then compared to the carrier signal (triangle wave) that is generated by itself. The following is the form of the signal that is processed to produce the SPWM signal.



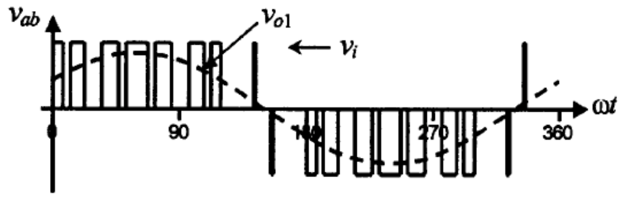
(a)



(b)



(c)



(d)

Fig. 8 a). Signal comparison, b). Signal forming at switch 1, c). Signal forming at switch 3, and d). SPWM result signal forming.

Because the sine signal compared is from the grid, the SPWM output signal will have the same phase and frequency as the grid. Thus synchronization control can be done using this method.

D. Low Pass Filter LC

The inverter output generally has a sine-shaped signal. But it is in the form of a multilevel signal that is sinus shaped. To produce a good sine signal a passive filter is needed which is also useful for reducing harmonics. The design of passive filters is designed based on 2nd order harmonics. The filter used is an LC filter. Inductance L is obtained from the power flow complex equation to ensure the current generated by the inverter flows to the grid to supply its load. The value of L is obtained at around 540uH through the following equation.

$$P = \frac{|V1|.|V2|}{X} .sin\,(\delta 1 - \delta 2) \qquad (2)$$

$$X = 2\pi f L \qquad (3)$$

From the above equation, the L inductance value is obtained to filter the power flow. To cut harmonics, the capacitor is required provided that the X_C value is the same as X_L. Thus the value of C:

$$X = \frac{1}{2\pi f C} \qquad (4)$$

Obtained filter capacitance value of 18uF.

III. Result and Discussion

The circuit below is a combination of a push-pull converter, three-phase inverter, and control in each circuit. The simulation uses two PVs connected in series. Radiation and temperature are 1000W/m2 and 25°C respectively. The simulation using PSIM software and implementation devices can be seen in the picture below.

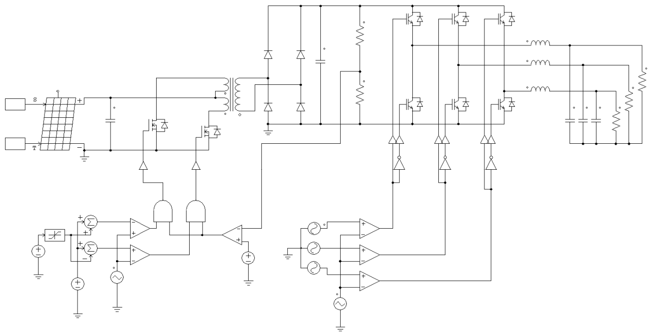


Fig. 9. Circuit scheme.

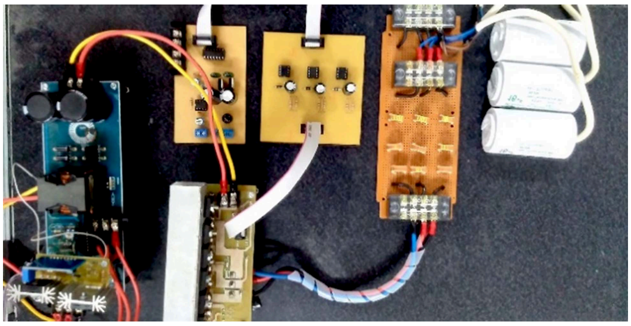


Fig. 10. Fabrication module of on-grid three phase inverter.

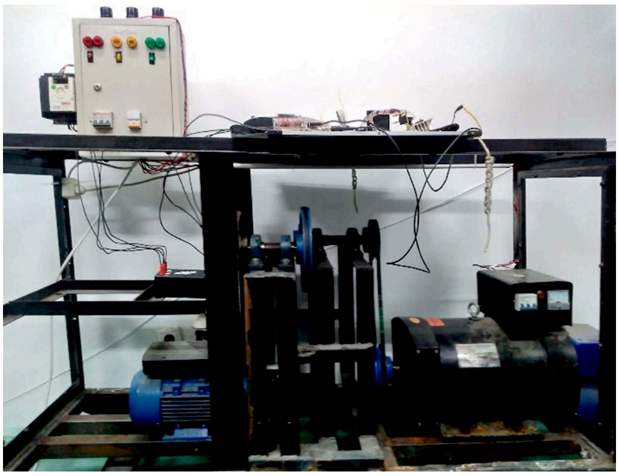


Fig. 11. The conditioned grid for testing the module .

In the picture below is a picture of the switching signal on the gate transistor and the push-pull converter output voltage. By using voltage-mode PWM SG3525, the push-pull converter output voltage can be adjusted according to the grid voltage used. In addition, it can maintain the output voltage due to input voltage fluctuations.

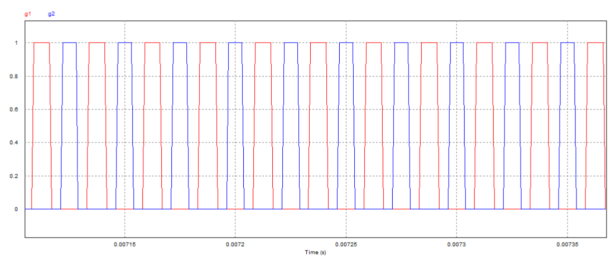


Fig. 12. Switching signal for Push-Pull Converter.

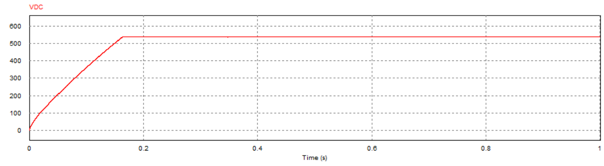


Fig. 13. Push-Pull Converter output voltage.

The three-phase inverter uses SPWM modulation which is obtained by comparing sine signals from three phase grids and triangular signals. The comparator results as the gate signal and the output of the three-phase inverter before being controlled can be seen in the figure below.

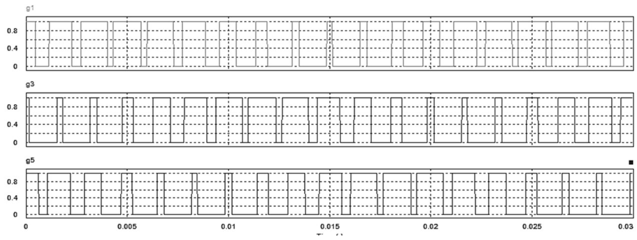


Fig. 14. The result signal of SPWM simulation.

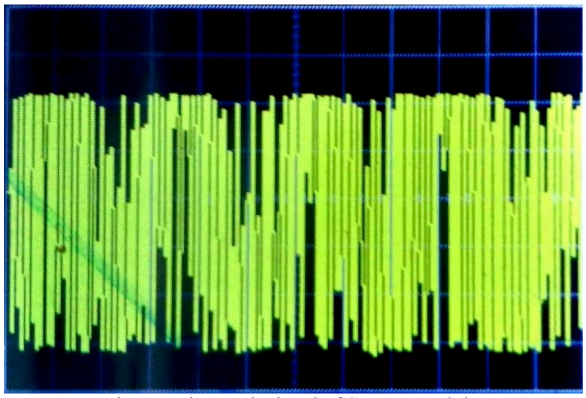


Fig. 15. The result signal of SPWM module.

The picture below is the result of simulation and testing using an oscilloscope. In this test, line-neutral THD of 1.18 will be muted using an LC filter.

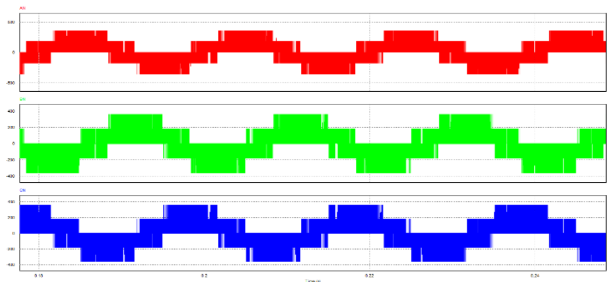


Fig. 16. The result of line – neutral inverter output simulation.

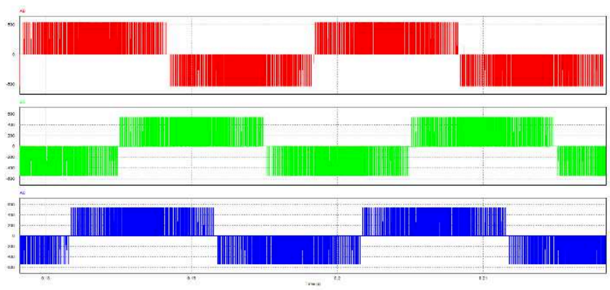


Fig. 17. The result of line – line inverter output simulation.

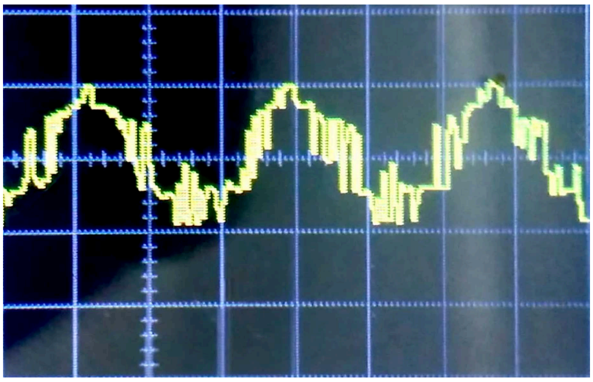


Fig. 18. The result of line – neutral inverter output module.

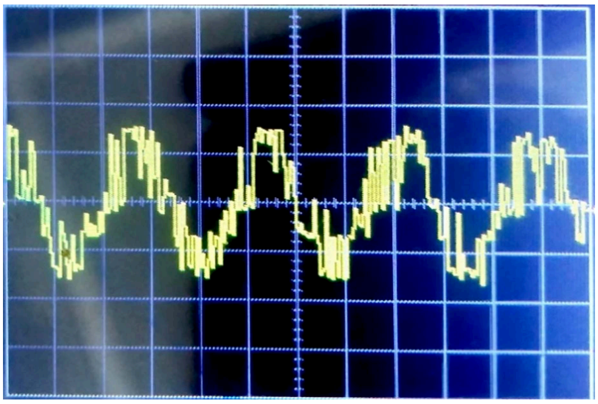


Fig. 19 The result of line – line inverter output module.

The three-phase inverter voltage is filtered using inductors 540uH and capacitors 16 uF. The result of line to neutral output voltage 216 V AC and line to line is 378 VAC. After the harmonics are muted using an LC filter, the result becomes 0.03. Because the harmonics after filtering are less than 5 percent, the inverter output can be connected to the grid. The resulting sine wave can be seen in the picture below.

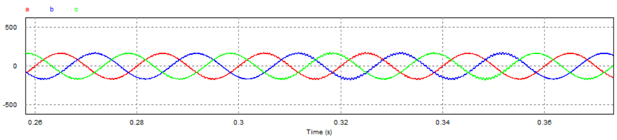


Fig. 20.Line – neutral of inverter output after filtered on simulation.

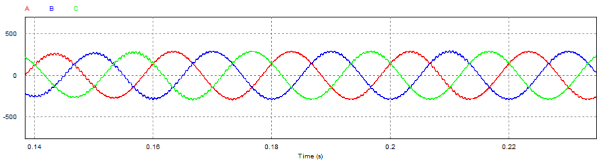


Fig. 21. Line – line of inverter output after filtered on simulation.

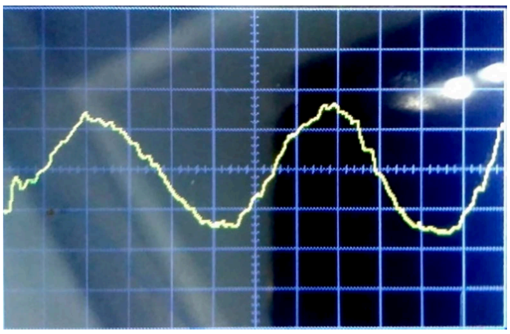


Fig. 22. The result of line – neutral inverter output module after filtered.

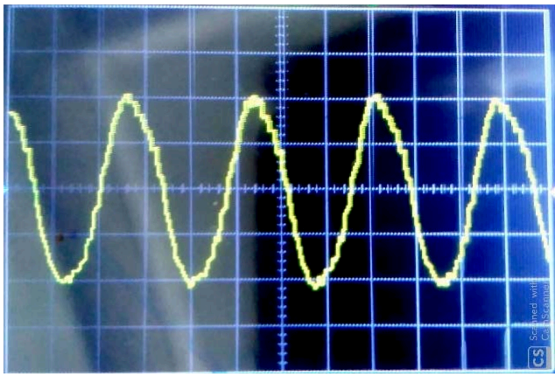


Fig. 23 The result of line – line inverter output module after filtered.

IV. CONCLUSION

This study can be used as a reference for designing 3-phase inverters in grid-connected solar panel systems. This can be proven by several things as follows:

1. The inverter's output voltage produces a 3-phase voltage from 1 DC source which is stepped up to obtain a 3-phase voltage.
2. The step-up method uses a push-pull converter to effectively increase the voltage from 60 VDC to 550 VDC
3. Grid synchronization to connect the inverter to the grid can use the SPWM method. Because the phase angle and frequency at the inverter output match the grid.
4. To produce a pure sine wave output, a low pass filter can be used to reduce the harmonics.

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